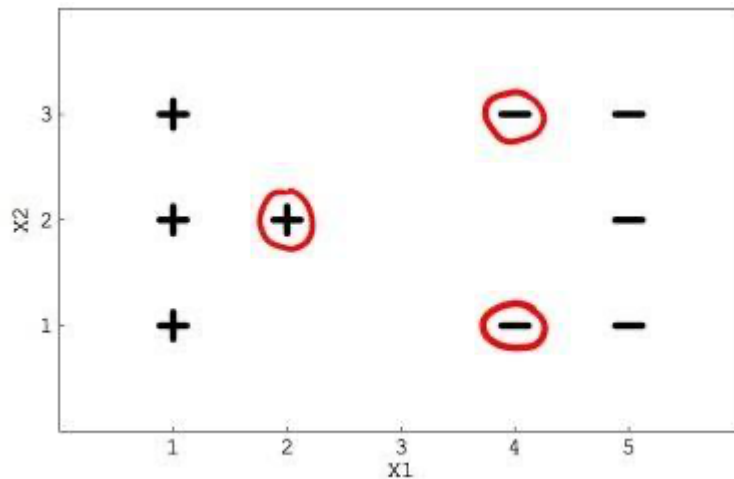


Tutorial Week 8 - SVM + GMM

SVM

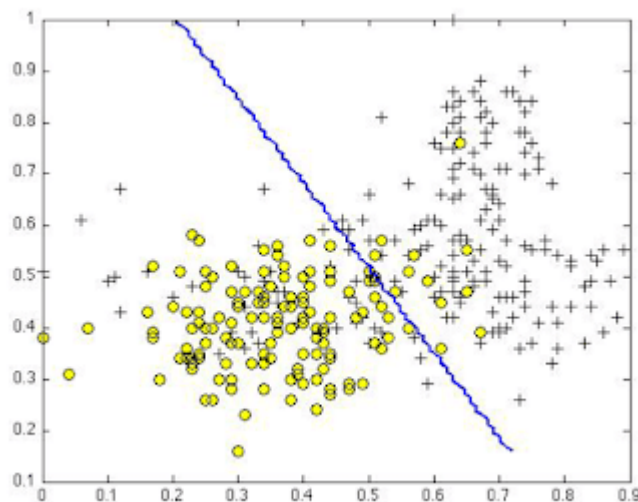
- Suppose you are using a Linear SVM classifier with 2 class classification problem. Now you have been given the following data in which some points are circled red that represent support vectors.



Part A: If you remove the following, anyone red points from the data. Will the decision boundary change?

Part B: If you remove the non-red circled points from the data, the decision boundary will change?

- Suppose you have trained an SVM classifier with a Gaussian kernel, and it learned the following decision boundary on the training set:

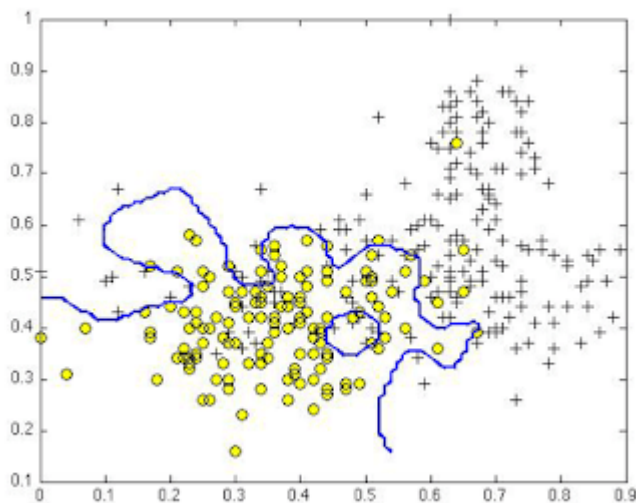


You suspect that the SVM is underfitting your dataset. Should you try increasing or decreasing C ?
Increasing or decreasing σ^2 ?

- ☐ It would be reasonable to try decreasing C . It would also be reasonable to try increasing σ^2 .

- ☐ It would be reasonable to try decreasing C . It would also be reasonable to try decreasing σ^2 .
- ☐ It would be reasonable to try increasing C . It would also be reasonable to try decreasing σ^2 .
- ☐ It would be reasonable to try increasing C . It would also be reasonable to try increasing σ^2 .

2. Suppose you have trained an SVM classifier with a Gaussian kernel, and it learned the following decision boundary on the training set:



When you measure the SVM's performance on a cross validation set, it does poorly. Should you try increasing or decreasing C ? Increasing or decreasing σ^2 ?

- ☐ It would be reasonable to try decreasing C . It would also be reasonable to try increasing σ^2 .
- ☐ It would be reasonable to try decreasing C . It would also be reasonable to try decreasing σ^2 .
- ☐ It would be reasonable to try increasing C . It would also be reasonable to try decreasing σ^2 .
- ☐ It would be reasonable to try increasing C . It would also be reasonable to try increasing σ^2 .

3. Suppose you have a dataset with $n = 10$ features and $m = 5000$ examples.

After training your logistic regression classifier with gradient descent, you find that it has underfit the training set and does not achieve the desired performance on the training or cross validation sets.

Which of the following might be promising steps to take? Check all that apply.

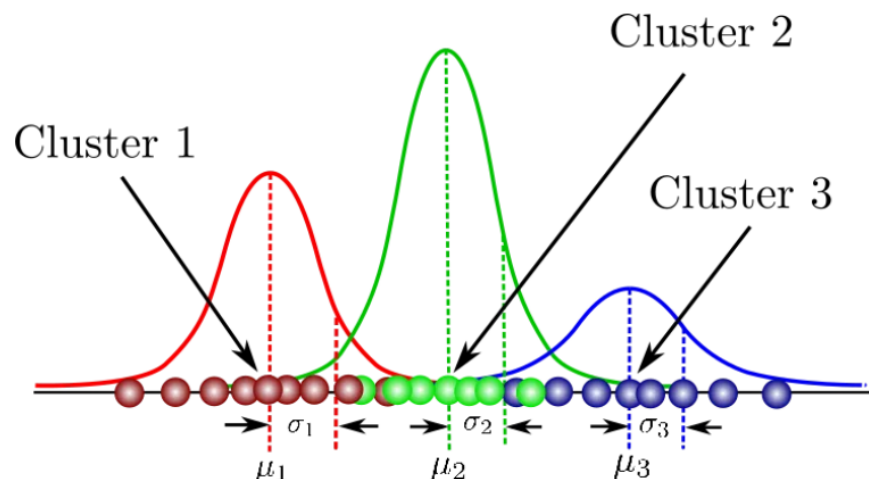
- ☐ Use an SVM with a Gaussian Kernel.

- ☐ Create / add new polynomial features.
 - ☐ Use an SVM with a linear kernel, without introducing new features.
 - ☐ Try using a neural network with a large number of hidden units.
 - ☐ Reduce the number of example in the training set.
4. Why is it important to perform feature normalization before using the Gaussian kernel?
 5. When the C parameter is set to infinite, which of the following holds true?
 - A) The optimal hyperplane, if exists, will be the one that completely separates the data
 - B) The soft-margin classifier will separate the data
 - C) None of the above
 6. The effectiveness of an SVM depends upon _____.
 - A) Selection of Kernel trick
 - B) Kernel Parameters
 - C) Soft Margin Parameter C
 - D) All of the above

GMM

Please identify if these statements are either True or False.

1. Of the clustering algorithms covered in class, Gaussian Mixture Models used for clustering always outperforms k-means and hierarchical clustering.
2. For the below Gaussian Mixture Model, answer the questions.



- I. How many Gaussian functions are depicted in the figure?

Gaussian distribution function is given by:

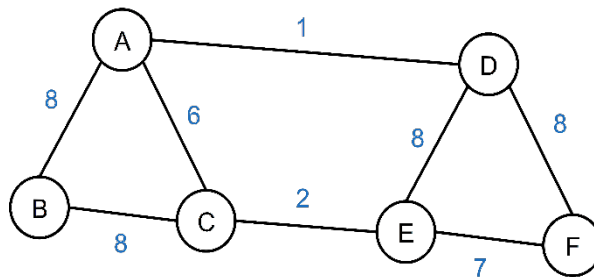
$$\mathcal{N}(\mathbf{x}|\mu, \Sigma) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$$

Where $\mathbf{x}$ represents our data points, D is the number of dimensions of each data point.

- II. What are the parameters to be estimated?
- III. If we have a dataset comprised of $N = 1000$ three-dimensional points ($D = 3$), then $\mathbf{X}$ will be a 1000×3 matrix. What will be the dimensions of the parameters?

Spectral Clustering

1. Consider the following graph with vertices A, B, C, D, E, F and 8 weights (edges).



2. Calculate the adjacency matrix.
3. Calculate the Laplacian Matrix
4. Normalize the graph Laplacian

$$L_{\text{sym}} = I - D^{-1/2} A D^{-1/2}$$

$$D^{-1/2} = \begin{bmatrix} 1/\sqrt{13} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/\sqrt{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/\sqrt{16} & 0 & 0 & 0 \end{bmatrix}$$